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MODEL SELECTION AND VALIDATION REPORT ( 2012_2016_biomass model )
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## OBSERVATIONS:801

## SANITY: OK − all models in the selection process passed sanity checks.

## SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1_with_anisotropy

## MODEL SUMMARY ( 2012_2016_biomass model )

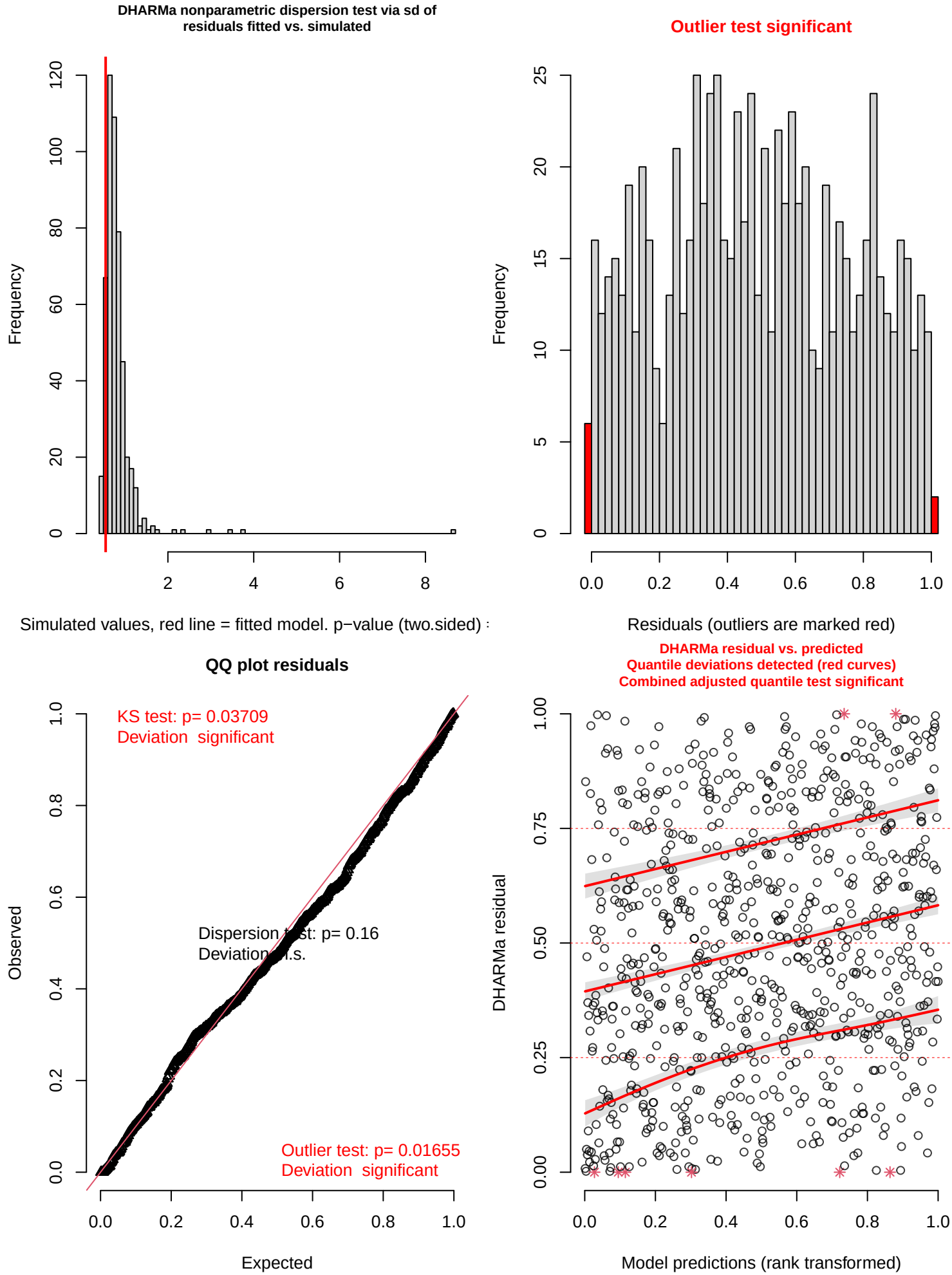
Spatiotemporal model fit by REML ['sdmTMB']
Formula: fixed_formula
Time column: year
Family: lognormal(link = 'log')

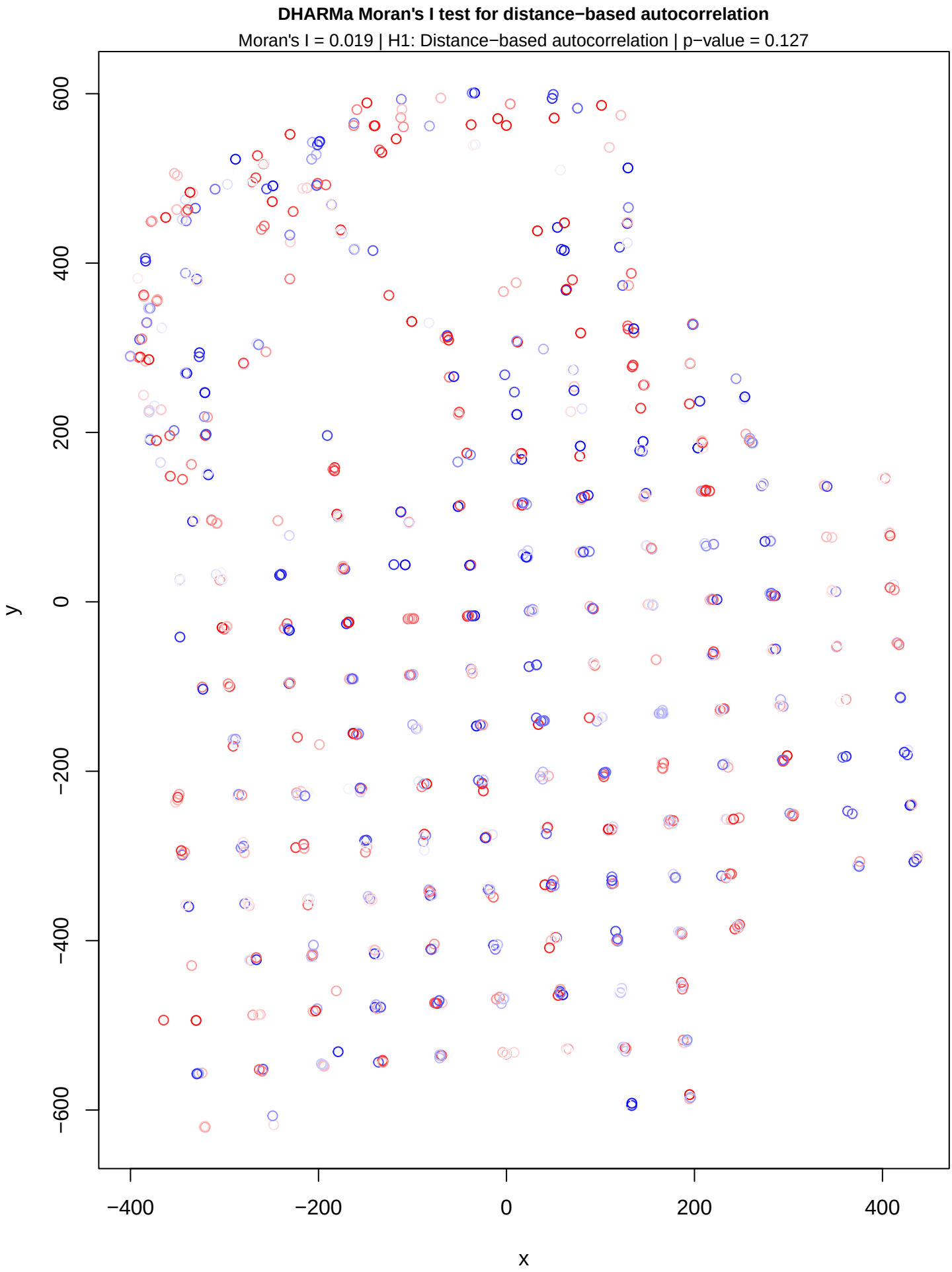
Conditional model:
              coef.est coef.se
(Intercept)       7.15    0.21
kde_fric          0.08    0.04
kde_feve         −0.11    0.04
tric              0.07    0.04
teve             −0.30    0.04
sbt_mean          0.26    0.09
sbt_sd            0.07    0.06
sst_sd            0.13    0.06
sic_mean         −0.13    0.09
log_chla_mean     0.00    0.04
depth            −0.06    0.05

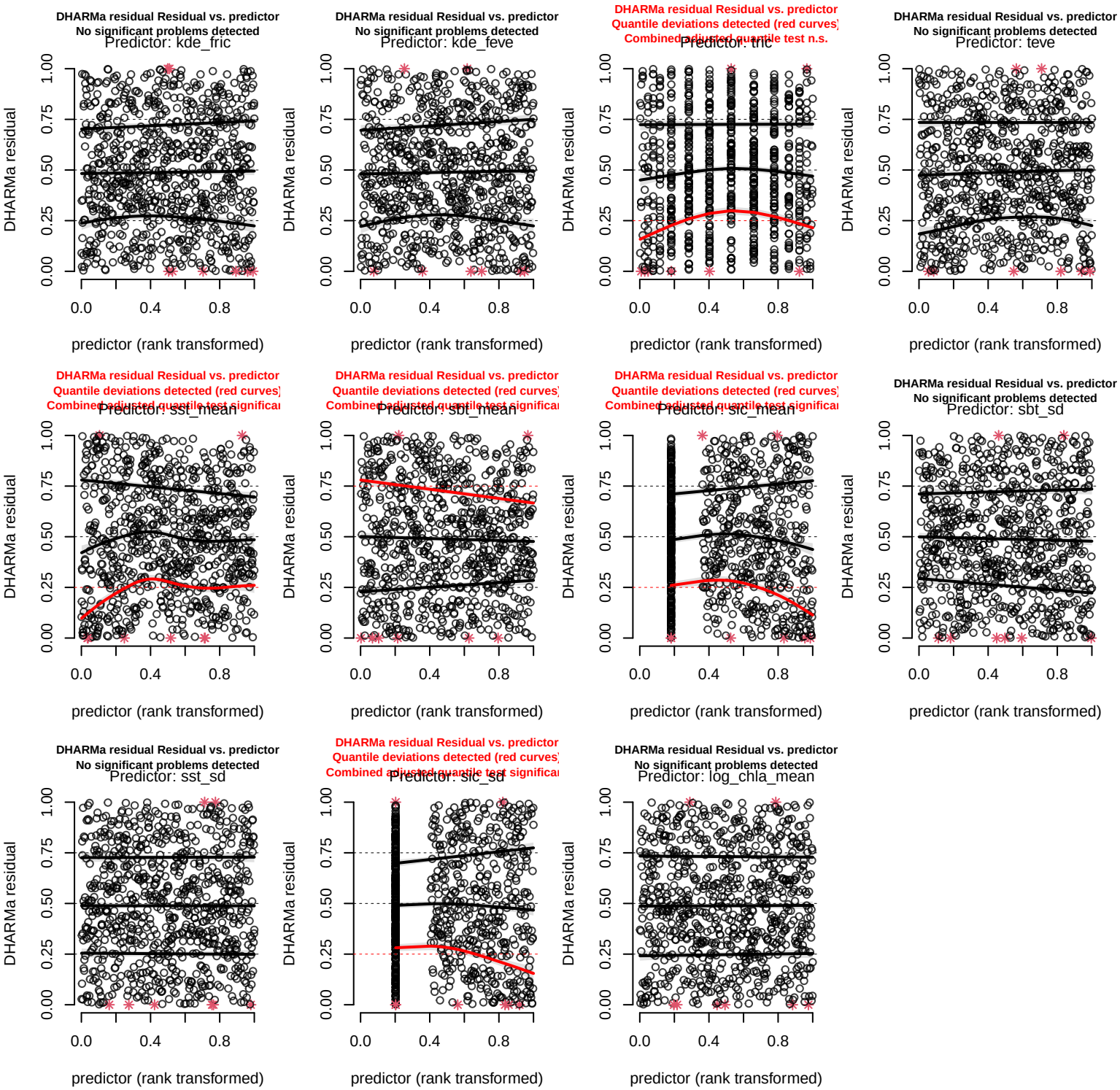
Dispersion parameter: 0.80
Spatiotemporal AR1 correlation (rho): 0.75
Matérn anisotropic range (spatiotemporal): 135.1 to 798.7 at 37 deg.
Spatiotemporal marginal AR1 SD: 0.69
REML criterion at convergence: 6536.499

See ?tidy.sdmTMB to extract these values as a data frame.
See ?plot_anisotropy to plot the anisotropic range.

## SELECTION STATISTICS
                                                   df      AIC   delta_AIC
spatiotemporal_random_fields_ar1_with_anisotropy 17 13107.00   6.3774832
spatiotemporal_random_fields_ar1                 15 13113.38  26.1646669
spatiotemporal_random_fields_iid_with_anisotropy 16 13139.54   5.5515234
spatial_random_fields_with_anisotropy            16 13145.09   3.6009728
spatial_random_fields                            14 13148.69   0.1140117
spatiotemporal_random_fields_iid                 14 13148.81 123.7741264
error_only                                       12 13272.58          NA
```


DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated
Frequency
Simulated values, red line = fitted model. p−value (two.sided) :
Outlier test significant
Frequency
Residuals (outliers are marked red)
QQ plot residuals
KS test: p= 0.03709
Deviation significant
Dispersion test: p= 0.16
Deviation n.s.
Outlier test: p= 0.01655
Deviation significant
Observed
Expected
DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
Model predictions (rank transformed)


DHARMa Moran's I test for distance−based autocorrelation
Moran's I = 0.019 | H1: Distance−based autocorrelation | p−value = 0.127
600
400
200
0
−200
−400
−600
y
−400
−200
0
200
400
x


DHARMa residual Residual vs. predictor
No significant problems detected
Predictor: kde_fric
DHARMa residual Residual vs. predictor
No significant problems detected
Predictor: kde_feve
DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.
Predictor: tric
DHARMa residual Residual vs. predictor
No significant problems detected
Predictor: teve
DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significa
Predictor: sst_mean
DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significa
Predictor: sbt_mean
DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significa
Predictor: sic_mean
DHARMa residual Residual vs. predictor
No significant problems detected
Predictor: sbt_sd
DHARMa residual Residual vs. predictor
No significant problems detected
Predictor: sst_sd
DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significa
Predictor: sic_sd
DHARMa residual Residual vs. predictor
No significant problems detected
Predictor: log_chla_mean
DHARMa residual
predictor (rank transformed)
0.00
0.25
0.50
0.75
1.00
0.0
0.4
0.8